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Producer is running. Semaphore full (address 0x3A) gets increased from 0 to 1.

EdSim51DI - Version 2.1.32 | testpreempt.hex

System Clock (MHz) 11.0529 1000 Update Freq.

SBUF

R/O	W/O	TH0	TL0	R7	B
0x00	0x00	0x05	0x00	0x01	0x00

RxD TxD 1 1 TMod 0x20 TCON 0xD0

SCON 0x50

pins bits TH1 TL1 0xFA 0xFC

0xFF 0xFF P3 0xFF 0xFF P2 0xFF 0xFF P1 0xFF 0xFF P0

PC 8051 PSW 1 0 0 0 1 0 0 0

Modify RAM

Data Memory

addr	0x00	0x00	value
0	0	1	2
1	3	0	0
2	0	1	0
3	0	1	0
4	0	1	0
5	0	1	0
6	0	1	0
7	0	1	0
8	0	1	0
9	0	1	0
A	0	1	0
B	0	1	0
C	0	1	0
D	0	1	0
E	0	1	0
F	0	1	0

Remove All Breakpo...

Time: 9ms 77us - Instructions: 5065

```
0065| MOV R6,A
0066| MOV 20H,R5
0068| SETB 0AFH
006A| MOV A,R7
006B| RRC A
006C| MOV 0AFH,C
006E*| INC 39H
0070*| INC 3AH
0072| SJMP 0A6H
0074| ORL 89H,#20H
0077| MOV 8DH,#0FAH
007A| MOV 98H,#50H
007D| SETB 8EH
007F| MOV 21H,#00H
0082| SETB 0AFH
0084| MOV A,3AH
0086| JB 0E7H,0FBH
0089| JZ 0F9H
008B| DEC 3AH
008D| MOV A,39H
```

P0.7 1 Display-select Decoder CS|DAC WR
P0.6 1 Keypad Column 2
P0.5 1 Keypad Column 1
P0.4 1 Keypad Column 0
P0.3 1 Keypad Row 3
P0.2 1 Keypad Row 2
P0.1 1 Keypad Row 1
P0.0 1 Keypad Row 0
P1.7 1 LED 7|Seg. dp|DAC DB7|LCD DB7
P1.6 1 LED 6|Seg. g|DAC DB6|LCD DB6
P1.5 1 LED 5|Seg. f|DAC DB5|LCD DB5
P1.4 1 LED 4|Seg. e|DAC DB4|LCD DB4
P1.3 1 LED 3|... d|..DB3|..DB3|.. RS
P1.2 1 LED 2|... c|..DB2|..DB2|LCD E
P1.1 1 LED 1|Seg. b|DAC DB1|LCD DB1
P1.0 1 LED 0|Seg. a|DAC DB0|LCD DB0
P2.7 1 SW 7|ADC DB7
P2.6 1 SW 6|ADC DB6
P2.5 1 SW 5|ADC DB5
P2.4 1 SW 4|ADC DB4
P2.3 1 SW 3|ADC DB3
P2.2 1 SW 2|ADC DB2
P2.1 1 SW 1|ADC DB1
P2.0 1 SW 0|ADC DB0
P3.7 1 ADC RD|Comparator Output
P3.6 1 ADC WR
P3.5 1 Motor Sensor
P3.4 1 Display-select Input 1
P3.3 1 AND Gate Output|Display-se..t 0
P3.2 1 ADC INTR
P3.1 1 Motor Control Bit 1|Ext. UART Rx
P3.0 1 Motor Control Bit 0|Ext. UART Tx

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System Clock (MHz) 11.0529 1000 Update Freq.

SBUF

R/O	W/O	TH0	TL0	R7	B
0x00	0x00	0x05	0x02	0x01	0x00

RxD TxD 1 1 TMod 0x20 TCON 0xD0

SCON 0x50

pins bits TH1 TL1 0xFA 0xFE

0xFF 0xFF P3 0xFF 0xFF P2 0xFF 0xFF P1 0xFF 0xFF P0

PC 8051 PSW 1 0 0 0 1 0 0 0

Modify RAM

Data Memory

addr	0x00	0x00	value
0	0	1	2
1	3	0	0
2	0	1	0
3	0	1	0
4	0	1	0
5	0	1	0
6	0	1	0
7	0	1	0
8	0	1	0
9	0	1	0
A	0	1	0
B	0	1	0
C	0	1	0
D	0	1	0
E	0	1	0
F	0	1	0

Remove All Breakpo...

Executed 0x0070: INC 3AH | Time: 9ms 80us

```
006E*| INC 39H
0070*| INC 3AH
0072| SJMP 0A6H
0074| ORL 89H,#20H
0077| MOV 8DH,#0FAH
007A| MOV 98H,#50H
007D| SETB 8EH
007F| MOV 21H,#00H
0082| SETB 0AFH
0084| MOV A,3AH
0086| JB 0E7H,0FBH
0089| JZ 0F9H
008B| DEC 3AH
008D| MOV A,39H
008F| JB 0E7H,0FBH
0092| JZ 0F9H
0094| DEC 39H
0096| MOV R7,#01H
0098| JBC 0AFH,02H
009B| MOV R7,#00H
009D| CLR 0AFH
```

P0.7 1
P0.6 1
P0.5 1
P0.4 1
P0.3 1
P0.2 1
P0.1 1
P0.0 1
P1.7 1
P1.6 1
P1.5 1
P1.4 1
P1.3 1
P1.2 1
P1.1 1
P1.0 1
P2.7 1
P2.6 1
P2.5 1
P2.4 1
P2.3 1
P2.2 1
P2.1 1
P2.0 1
P3.7 1
P3.6 1
P3.5 1
P3.4 1
P3.3 1
P3.2 1
P3.1 1
P3.0 1

Consumer is running. Semaphore empty (address 0x3B) gets increased from 0 to 1.

EdSim51DI - Version 2.1.32 | testpreempt.hex

System Clock (MHz) 11.0529 1000 Update Freq.

SBUF

R/O	W/O	TH0	TL0	R7	B
0x00	0x41	0x05	0x1B	0x01	0x00

R/D T/D

R/D	T/D	TMOD	PCON
1	0	0x20	0x00

SCON 0x50 TCON 0xD0

pins bits

pins	bits	TH1	TL1
0xFF	0xFF	P3	0xFA
0xFF	0xFF	P2	0xFB
0xFF	0xFF	P1	0xFC
0xFF	0xFF	P0	0xFD

PC 0x00C2 PSW 1 0 0 0 1 0 0 0

8051

Data Memory

addr	0x00	0x01	value
0	00	30	00
1	00	00	01
2	00	01	00
3	00	01	01
4	00	01	01
5	00	31	3C
6	00	00	00
7	00	01	01
8	00	00	00
9	00	00	00
A	00	00	00
B	00	00	00
C	00	00	00
D	00	00	00
E	00	00	00
F	00	00	00

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Remove All Breakpo...

RST Step Run New Load Save Copy Paste

Time: 18ms 1us - Instructions: 10028

```
00B6| KLC A
00B7| SUBB A,0E0H
00B9| MOV R6,A
00BA| MOV 21H,R5
00BC| SETB 0AFH
00BE| MOV A,R7
00BF| RRC A
00C0| MOV 0AFH,C
00C2*| INC 39H
00C4*| INC 3BH
00C6| JBC 99H,0BBH
00C9| SJMP 0FBH
00CB| MOV 3AH,#00H
00CE| MOV 39H,#01H
00D1| MOV 3BH,#03H
00D4| MOV DPTR,#0014H
00D7| LCALL 0125H
00DA| LJMP 0074H
00DD| LJMP 00E8H
00E0| RET
00E1| RET
```

P0.7 1 Display-select Decoder CS|DAC WR
P0.6 1 Keypad Column 2
P0.5 1 Keypad Column 1
P0.4 1 Keypad Column 0
P0.3 1 Keypad Row 3
P0.2 1 Keypad Row 2
P0.1 1 Keypad Row 1
P0.0 1 Keypad Row 0
P1.7 1 LED 7|Seg. dp|DAC DB7|LCD DB7
P1.6 1 LED 6|Seg. g|DAC DB6|LCD DB6
P1.5 1 LED 5|Seg. f|DAC DB5|LCD DB5
P1.4 1 LED 4|Seg. e|DAC DB4|LCD DB4
P1.3 1 LED 3|... d|.DB3|.DB3|.RS
P1.2 1 LED 2|... c|.DB2|.DB2|LCD E
P1.1 1 LED 1|Seg. b|DAC DB1|LCD DB1
P1.0 1 LED 0|Seg. a|DAC DB0|LCD DB0
P2.7 1 SW 7|ADC DB7
P2.6 1 SW 6|ADC DB6
P2.5 1 SW 5|ADC DB5
P2.4 1 SW 4|ADC DB4
P2.3 1 SW 3|ADC DB3
P2.2 1 SW 2|ADC DB2
P2.1 1 SW 1|ADC DB1
P2.0 1 SW 0|ADC DB0
P3.7 1 ADC RD|Comparator Output
P3.6 1 ADC WR
P3.5 1 Motor Sensor
P3.4 1 Display-select Input 1
P3.3 1 AND Gate Output|Display-se..t 0
P3.2 1 ADC INTR
P3.1 1 Motor Control Bit 1|Ext. UART Rx
P3.0 1 Motor Control Bit 0|Ext. UART Tx

EdSim51DI - Version 2.1.32 | testpreempt.hex

System Clock (MHz) 11.0529 1000 Update Freq.

SBUF

R/O	W/O	TH0	TL0	R7	B
0x00	0x42	0x3C	0x12	0x01	0x00

R/D T/D

R/D	T/D	TMOD	PCON
1	1	0x20	0x00

SCON 0x50 TCON 0xD0

pins bits

pins	bits	TH1	TL1
0xFF	0xFF	P3	0xFA
0xFF	0xFF	P2	0xFB
0xFF	0xFF	P1	0xFC
0xFF	0xFF	P0	0xFD

PC 0x00C2 PSW 1 0 0 0 1 0 0 0

8051

Data Memory

addr	0x00	0x01	value
0	00	30	00
1	00	00	01
2	00	00	00
3	00	00	00
4	00	00	00
5	00	00	00
6	00	00	00
7	00	00	00
8	00	00	00
9	00	00	00
A	00	00	00
B	00	00	00
C	00	00	00
D	00	00	00
E	00	00	00
F	00	00	00

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Remove All Breakpo...

RST Step Run New Load Save Copy Paste

Time: 19ms 902us - Instructions: 10914

```
00B6| KLC A
00B7| SUBB A,0E0H
00B9| MOV R6,A
00BA| MOV 21H,R5
00BC| SETB 0AFH
00BE| MOV A,R7
00BF| RRC A
00C0| MOV 0AFH,C
00C2*| INC 39H
00C4*| INC 3BH
00C6| JBC 99H,0BBH
00C9| SJMP 0FBH
00CB| MOV 3AH,#00H
00CE| MOV 39H,#01H
00D1| MOV 3BH,#03H
00D4| MOV DPTR,#0014H
00D7| LCALL 0125H
00DA| LJMP 0074H
00DD| LJMP 00E8H
00E0| RET
00E1| RET
```

P0.7 1 Display-select Decoder CS|DAC WR
P0.6 1 Keypad Column 2
P0.5 1 Keypad Column 1
P0.4 1 Keypad Column 0
P0.3 1 Keypad Row 3
P0.2 1 Keypad Row 2
P0.1 1 Keypad Row 1
P0.0 1 Keypad Row 0
P1.7 1 LED 7|Seg. dp|DAC DB7|LCD DB7
P1.6 1 LED 6|Seg. g|DAC DB6|LCD DB6
P1.5 1 LED 5|Seg. f|DAC DB5|LCD DB5
P1.4 1 LED 4|Seg. e|DAC DB4|LCD DB4
P1.3 1 LED 3|... d|.DB3|.DB3|.RS
P1.2 1 LED 2|... c|.DB2|.DB2|LCD E
P1.1 1 LED 1|Seg. b|DAC DB1|LCD DB1
P1.0 1 LED 0|Seg. a|DAC DB0|LCD DB0
P2.7 1 SW 7|ADC DB7
P2.6 1 SW 6|ADC DB6
P2.5 1 SW 5|ADC DB5
P2.4 1 SW 4|ADC DB4
P2.3 1 SW 3|ADC DB3
P2.2 1 SW 2|ADC DB2
P2.1 1 SW 1|ADC DB1
P2.0 1 SW 0|ADC DB0
P3.7 1 ADC RD|Comparator Output
P3.6 1 ADC WR
P3.5 1 Motor Sensor
P3.4 1 Display-select Input 1
P3.3 1 AND Gate Output|Display-se..t 0
P3.2 1 ADC INTR
P3.1 1 Motor Control Bit 1|Ext. UART Rx
P3.0 1 Motor Control Bit 0|Ext. UART Tx